

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

CHAPTER – II

IDEATION PHASE

2.1-Problem Statement:

Problem Statement

The dietary habits of college students play a significant role in their overall health, academic performance, and quality of life. However, many students struggle to maintain a balanced diet due to factors such as busy academic schedules, financial limitations, stress, irregular meal timings, and the easy availability of fast food and processed foods. These challenges often result in unhealthy eating patterns, inadequate nutritional intake, and lifestyle-related health issues such as obesity, fatigue, poor concentration, and reduced physical fitness.

Educational institutions and health professionals require a better understanding of students' food choices and nutritional behaviors to develop effective health and wellness programs. Although colleges often collect survey data related to students' dietary habits, exercise routines, and health perceptions, the information is usually stored in large datasets that are difficult to interpret using traditional methods. Without proper analysis and visualization, it becomes challenging to identify meaningful trends, compare different student groups, and understand the factors influencing dietary decisions.

The project "Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study" addresses this challenge by transforming raw dietary survey data into interactive and easy-to-understand visualizations using Tableau. The project analyzes key aspects such as demographic distribution, daily calorie intake, meal frequency, favorite cuisine preferences, vegetable consumption, dining-out habits, exercise frequency, healthy eating perception, and overall health ratings. These visualizations enable users to identify patterns, correlations, and trends that may not be apparent from raw data alone.

By providing interactive dashboards and stories, the project supports data-driven decision-making for students, educators, nutritionists, and college administrators. Users can explore different aspects of dietary behavior, compare food choices across various demographic groups, and evaluate the relationship between eating habits, physical activity, and health

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outcomes. The insights generated through this analysis help identify areas where students may require nutritional guidance or lifestyle improvements.

The project also aims to recommend practical dietary strategies that encourage healthier eating habits and promote nutritional awareness among college students. These recommendations can assist educational institutions in designing effective wellness campaigns, nutrition education programs, and healthy cafeteria initiatives. Ultimately, the project demonstrates how Tableau can be used as a powerful data visualization tool to convert complex dietary data into meaningful insights, supporting better health, informed decision-making, and improved student well-being.

